StructRAG: HTML Structure-Aware Retrieval for Document-Centric Question Answering

Research Paper Outline & Positioning

1. Title & Abstract

Title: "StructRAG: Leveraging HTML DOM Hierarchy for Efficient Structure-Aware Retrieval-Augmented Generation"

Abstract: Current RAG systems face a fundamental trade-off: simple chunking methods lose critical structural information, while graph-based approaches require expensive LLM-based extraction. We propose StructRAG, a novel approach that preserves document hierarchy through HTML DOM structure without additional extraction costs. Our method achieves X% improvement in retrieval accuracy on structured documents while reducing computational overhead by Y% compared to GraphRAG.

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2. Problem Statement & Research Gap (Updated)

Current State of RAG Systems:

- **Flat Chunking Methods**: Fast but lose structural relationships
- **GraphRAG**: Preserves relationships but requires expensive LLM extraction
- **HtmlRAG (WWW 2025)**: Preserves HTML structure but lacks graph traversal capabilities
- **Document-Centric Queries**: Limited multi-hop reasoning over document structure

Identified Research Gap (Post-HtmlRAG):

> "While HtmlRAG preserves HTML structure as format, no existing approach treats DOM hierarchy as a traversable knowledge graph for multi-hop structural reasoning"

Updated Research Questions:

1. Can DOM hierarchy serve as a zero-cost knowledge graph for multi-hop document traversal?
2. How does graph-based DOM traversal compare to HtmlRAG's block-level processing?
3. What is the optimal integration of structural graph traversal with semantic similarity?
4. Can DOM graph traversal enhance HtmlRAG's format-preserving approach?

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3. Novel Technical Contributions

Core Innovation 1: Zero-Cost Hierarchical Graph Construction

```python
# Novel: Direct DOM-to-graph mapping without LLM extraction
for sibling in heading.next_siblings:
    if sibling.name in ['h1', 'h2', 'h3', 'h4', 'h5', 'h6']:
        break  # Natural hierarchy boundaries
```

Contribution: First approach to use HTML DOM as implicit knowledge graph for RAG

Core Innovation 2: Hybrid Semantic-Structural Scoring

```python
hybrid_score = α * semantic_similarity(query, chunk) +
               β * structural_coherence(chunk, context) +
               γ * hierarchy_relevance(chunk.level, query_type)
```

Contribution: Novel multi-signal ranking combining vector similarity with structural features

Core Innovation 3: Structure-Preserving Embeddings

```python
chunk_embedding = embed(content + structural_context)
# where structural_context = f"<{html_tag} level={hierarchy_level}>{heading}</{html_tag}>"
```

Contribution: First method to encode structural metadata directly in vector representations

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4. Methodology

System Architecture:

1. **HTML Structure Extraction**: DOM parsing → hierarchical node graph
2. **Structure-Aware Chunking**: Preserve heading boundaries + context
3. **Hybrid Embedding**: Content + structural metadata → vectors
4. **Multi-Signal Retrieval**: Semantic + structural + positional ranking
5. **Context-Aware Generation**: Structure-informed prompt construction

Experimental Design (Updated):

- **Datasets**: Wikipedia, MDN Web Docs, Enterprise documentation
- **Baselines**: Traditional RAG, **HtmlRAG**, GraphRAG, RAPTOR, HyDE

- **Metrics**: Retrieval precision/recall, answer quality (BLEU, ROUGE), computational efficiency, structural coherence
- **Key Comparisons**:
  - **vs HtmlRAG**: Graph traversal vs block-level processing
  - **vs GraphRAG**: DOM extraction vs LLM-based entity extraction
  - **vs Traditional RAG**: Structured vs unstructured retrieval
- **Ablation Studies**: Individual contribution of each structural signal, DOM traversal depth analysis

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5. Expected Results & Theoretical Framework

Hypothesis:

Document structure provides orthogonal signal to semantic similarity, improving retrieval for:

- **Hierarchical Queries**: "What are the subsections under X?"
- **Contextual Queries**: Questions requiring section-level understanding
- **Multi-hop Reasoning**: Following document flow and references

Theoretical Contributions:

1. **Structural Relevance Theory**: Formalize relationship between DOM hierarchy and information relevance
2. **Hybrid Ranking Framework**: Mathematical model for combining semantic and structural signals
3. **Computational Complexity Analysis**: Prove $O(n)$ vs $O(n^2)$ advantage over graph extraction methods

Expected Performance:

- **Accuracy**: 15-25% improvement on structured document QA tasks
- **Efficiency**: 60-80% reduction in preprocessing time vs GraphRAG
- **Scalability**: Linear scaling with document size (vs quadratic for graph methods)

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6. Positioning in Literature

Related Work Comparison:

Approach	Structure Preservation	Extraction Cost	Multi-hop Reasoning	Graph Traversal	HTML Native
Traditional RAG	❌ None	✅ Zero	❌ Limited	❌ None	❌ No
GraphRAG	✅ Full	❌ High (LLM calls)	✅ Excellent	✅ Advanced	❌ No
RAPTOR	🟡 Hierarchical	🟡 Medium	🟡 Good	🟡 Tree-based	❌ No
HtmlRAG	✅ HTML Tags	✅ Zero	🟡 Limited	❌ None	✅ Yes
StructRAG (Ours)	✅ DOM Graph	✅ Zero	✅ DOM-based	✅ DOM Traversal	✅ Yes

Key Distinction from HtmlRAG:

HtmlRAG (WWW 2025) represents the most directly related work, but differs fundamentally in approach:

Aspect	HtmlRAG	StructRAG (Ours)
Core Philosophy	HTML as richer text format	DOM as implicit knowledge graph
Structure Usage	Preserves through HTML tags	Explicitly models as graph relationships
Retrieval Method	Embedding similarity on HTML blocks	Hybrid semantic + structural traversal
Multi-hop Reasoning	Limited to block-level context	Full DOM tree traversal capabilities
Graph Formalism	None - treats structure as metadata	Formal graph theory with traversal algorithms
Context Expansion	Block pruning and compression	Relationship-based context discovery

Complementary Positioning: StructRAG can enhance HtmlRAG's format-preserving approach by adding graph traversal capabilities for complex queries requiring structural reasoning.

Novel Positioning (Updated):

- **First** to formalize DOM hierarchy as traversable knowledge graph for RAG
- **Novel** graph-theoretic approach to HTML document relationships
- **Unique** zero-cost graph extraction vs expensive LLM-based methods (GraphRAG)
- **Complementary** to HtmlRAG's format-preserving approach with graph traversal capabilities

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7. Implementation & Evaluation Plan

Phase 1: Core System Development (Months 1-4)

- Implement HTML structure extraction pipeline
- Develop hybrid ranking algorithms
- Create structure-aware embedding methods

Phase 2: Experimental Validation (Months 5-8)

- Benchmark on standard datasets (MS MARCO, Natural Questions)
- Create new HTML-specific evaluation datasets
- Conduct extensive ablation studies

Phase 3: Real-world Deployment (Months 9-12)

- Enterprise documentation system deployment
- User study with software developers
- Performance optimization and scaling analysis

Phase 4: Publication & Dissemination (Months 11-12)

- Paper submission to top-tier venues (ACL, EMNLP, AAAI)
- Open-source implementation release
- Industry whitepaper and tech talks

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8. Target Venues

Primary Targets:

- **ACL/EMNLP**: Strong NLP focus, receptive to RAG innovations
- **AAAI**: AI systems emphasis, good for hybrid approaches
- **WWW/SIGIR**: Web document focus aligns perfectly

Secondary Targets:

- **ICLR**: Machine learning methodology focus
- **ICML**: If theoretical contributions are strong
- **Industry Conferences**: KDD, WSDM for practical impact

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9. Potential Impact & Significance

Academic Impact:

- **New Research Direction**: Structure-aware RAG systems
- **Theoretical Framework**: Formal treatment of document hierarchy in retrieval
- **Benchmark Datasets**: HTML-specific evaluation resources

Industrial Impact:

- **Documentation Systems**: Improved developer experience
- **Enterprise Knowledge Bases**: Better internal search
- **Educational Platforms**: Enhanced learning material navigation
- **Legal/Medical Documents**: Structure-sensitive information retrieval

Societal Benefits:

- More accurate information retrieval from authoritative sources
- Reduced computational resources for knowledge access systems
- Better preservation of author-intended document structure

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10. Risk Analysis & Mitigation

Technical Risks:

- **HTML Quality Dependence**: Mitigate with robust parsing + fallback methods
- **Limited Applicability**: Address with multi-format support (Markdown, XML)
- **Scalability Concerns**: Optimize with efficient data structures

Validation Risks:

- **Dataset Bias**: Use diverse, real-world HTML sources
- **Baseline Fairness**: Ensure comparable computational budgets
- **Evaluation Metrics**: Develop structure-aware evaluation measures

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11. Timeline & Milestones

Year 1:

- Q1: Literature review, initial implementation
- Q2: Core algorithm development
- Q3: Preliminary experiments

- Q4: First paper submission

Year 2:

- Q1: Major venue publication
- Q2: System scaling and optimization
- Q3: Industry collaboration and deployment
- Q4: Follow-up research and extensions

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12. Expected Publications

1. **Main Paper**: "StructRAG: HTML Structure-Aware Retrieval-Augmented Generation" (ACL/EMNLP)
2. **System Paper**: "Efficient Implementation of Structure-Aware RAG at Scale" (KDD)
3. **Analysis Paper**: "When Structure Matters: Analyzing the Role of Document Hierarchy in Information Retrieval" (SIGIR)
4. **Survey Paper**: "Beyond Flat Text: A Survey of Structure-Aware Retrieval Methods" (AI Survey Journal)

This research has the potential to establish a new paradigm in RAG systems and open multiple follow-up research directions!